

Model Question Paper-I with effect from 2025

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1BMATS101

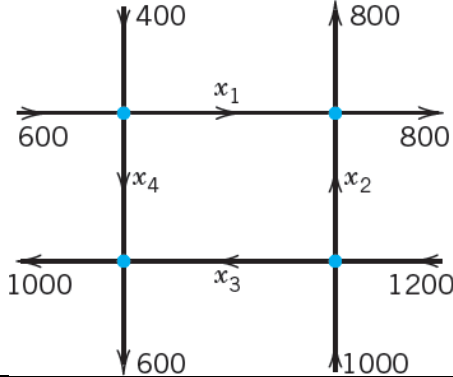
First Semester B.E./B.Tech. Degree Examination Calculus & Linear Algebra

TIME: 03Hours

Max.Marks:100

- Note: 1. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**
 2. VTU Formula Hand Book is Permitted
 3. M: Marks, L: Bloom's level, C: Course outcomes

Module-1			M	L	C
Q 1.	a	Show that $u_x + u_y = u$, if $u = \frac{e^{x+y}}{e^x + e^y}$.	6	L2	1
	b	If $u = x + 3y^2 - z^3$, $v = 4x^2yz$, $w = 2z^2 - xy$, find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ at (1, -1, 0).	7	L2	1
	c	Find the extreme values of the function $f(x, y) = xy + \frac{a^3}{x} + \frac{a^3}{y}$.	7	L3	1
OR					
Q 2.	a	If $V = f(r, s, t)$ and $r = \frac{x}{y}$, $s = \frac{y}{z}$, $t = \frac{z}{x}$ show that $x \frac{\partial V}{\partial x} + y \frac{\partial V}{\partial y} + z \frac{\partial V}{\partial z} = 0$.	6	L2	1
	B	If $u = \frac{2yz}{x}$, $v = \frac{3zx}{y}$, $w = \frac{4xy}{z}$ then find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$.	7	L2	1
	C	Apply Maclaurin's series, to expand $\cos x \cos y$ in powers of x and y up to second-degree terms.	7	L3	1
Module-2					
Q 3.	a	If $f = x^2yz$ and $g = xy - 3z^2$, calculate $\nabla(\nabla f \cdot \nabla g)$.	6	L2	1
	b	A vector field is given by $F = (6xy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - y)\hat{k}$. Show that the field is irrotational and hence find its scalar potential.	7	L2	1
	c	Express the vector $\vec{F} = x\hat{i} + 2y\hat{j} + yz\hat{k}$ in spherical polar coordinates.	7	L3	1
OR					
Q 4.	a	Find the directional derivative of $f(x, y, z) = 4e^{2x-y+z}$ at the point (1, 1, -1) in the direction towards the point (-3, 5, 6).	6	L2	1
	b	Find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$, where $F = \text{grad}(x^3 + y^3 + z^3 - 3xyz)$.	7	L2	1
	c	Express the vector $\vec{F} = 2x\hat{i} + 3y\hat{j} - z\hat{k}$ in cylindrical polar coordinates.	7	L3	1
Module-3					

Q 5.	A	Find the constant b if the rank of $\begin{bmatrix} 1 & 1 & -1 & 0 \\ 4 & 4 & -3 & 1 \\ b & 2 & 2 & 2 \\ 9 & 9 & b & 3 \end{bmatrix}$ is 3.	6	L2	2
	b	Find model matrix of $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ and verify its diagonalization.	7	L2	2
	c	Write the system of linear equations of the traffic flow in the net of one-way street directions as shown in the figure and find its solution. 	7	L3	2
OR					
Q 6.	a	Investigate the values of λ and μ so that the equations $2x + 3y + 5z = 9$ $7x + 3y - 2z = 8$ $2x + 3y + \lambda z = \mu$ have (i) no solution (ii) unique solution (iii) infinite number of solutions.	6	L2	2
	b	Apply Gauss Jordan method to approximate the solutions of the system $83x + 11y - 4z = 95$ $7x + 52y + 13z = 104$ by choosing initial solution $(0,0,0)$. Perform four iterations.	7	L2	2
	C	Determine the eigenvalues and corresponding eigenvectors for the matrix $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$.	7	L3	2
Module-4					
Q 7.	a	Verify whether $v = (1, -2, 5)$ in $\mathbb{R}^3$ is a linear combination of the vectors $u_1 = (1, 1, 1)$, $u_2 = (1, 2, 3)$ and $u_3 = (2, -1, 1)$.	6	L2	3
	b	Determine whether $W = \{(a, b, c) / a + b + c = 0\}$ is a subspace of R^3 or not?	7	L2	3
	c	Find the basis and dimension of the row space, column space and null space of the matrix $\begin{bmatrix} 1 & -1 & 1 & 3 & 2 \\ 2 & -1 & 1 & 5 & 1 \\ 0 & 1 & -1 & -1 & -3 \end{bmatrix}$	7	L2	3

OR					
Q 8.	a	Find the basis and dimension of the subspace W spanned by $(1, 2, 3), (2, 4, 6), (0, 1, 1)$.	6	L2	3
	b	Find the inner products $\langle v_1, v_2 \rangle, \langle v_1, v_3 \rangle$ and $\langle v_2, v_3 \rangle$ where $v_1 = (1, 1, 1, 1)$, $v_2 = (1, 2, 4, 5)$ $v_3 = (1, -3, -4, -2)$.	7	L2	3
	c	Find the coordinates of the vector $v = (1, -3, 2)$ with respect to the basis $S = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$.	7	L2	3
Module-5					
Q 9.	a	Verify whether the transformation $T: R^2 \rightarrow R^2$ which is defined by $T(x, y) = (3x + 4y, 10x - 4y + 3)$ is linear or not?	6	L2	3
	b	Prove that the transformation $F: R^2 \rightarrow R^2$ is singular and find its Kernal if the transformation $F(a, b) = (2a - 4b, 3a - 6b)$.	7	L2	3
	c	Find the rank and nullity of the transformation $T: R^3 \rightarrow R^3$ defined by $T(x, y, z) = (x + y, x - y, 2x + z)$.	7	L2	3
OR					
Q 10.	A	Check whether the transformation $T: V_1(R) \rightarrow V_3(R)$ defined by $T(x) = (x, x^2, x^3)$ is linear or not.	6	L2	3
	b	Consider the matrix $A = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix}$ which defines a linear operator on $\mathbb{R}^2$. Find the matrix of the linear transformation relative to the basis $S = \{u_1, u_2\} = \left\{ \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 3 \\ -7 \end{bmatrix} \right\}$.	7	L2	3
		Let F be the linear transformation defined on a vector space R^2 through $F(x, y) = (2x + y, 3x + 2y)$, show that F is invertible and hence find F^{-1} .	7	L2	3

CBCS SCHEME

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First Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026 Calculus and Linear Algebra : CSE Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.
3. VTU Handbook is permitted.*

Module - 1				M	L	C
Q.1	a.	Find the total derivative when $u = x^3y^2 + x^2y^3$ where $x = at^2, y = 2at$		6	L2	CO1
	b.	If $u = \log(\tan x + \tan y + \tan z)$ then, show that $\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2$		7	L2	CO1
	c.	Expand $e^x \sin y$ in the powers of x and y as far as terms of third term using Maclaurin's series.		7	L3	CO1
OR						
Q.2	a.	If $u = x + y + z, v = x^2 + y^2 + z^2, w = xy + yz + zx$ Evaluate the Jacobian of (u, v, w) with respect to (x, y, z) .		6	L2	CO1
	b.	If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$		7	L2	CO1
	c.	Examine the function for the extreme values, given $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$		7	L3	CO1
Module - 2						
Q.3	a.	Find the directional derivative of $\phi = x^2yz + 4xz^2$ at the point $(1, -2, -1)$ in the direction $2\hat{i} - \hat{j} - 2\hat{k}$.		6	L2	CO1
	b.	Show that $\vec{F} = (y+z)\hat{i} + (z+x)\hat{j} + (x+y)\hat{k}$ is irrotational and find the scalar potential.		7	L2	CO1
	c.	With usual notation, Prove that cylindrical polar coordinate system is orthogonal.		7	L3	CO1
OR						
Q.4	a.	Given $\vec{F} = \nabla(xy^3z^2)$, Find $\text{div}(\vec{F})$ and $\text{curl}(\vec{F})$ at $(1, -1, 1)$.		6	L2	CO1
	b.	Show that $\vec{F} = \frac{x\hat{i} + y\hat{j}}{x^2 + y^2}$ is both solenoidal and irrotational.		7	L2	CO1
	c.	Express $2y\hat{i} - 3z\hat{j} + 3x\hat{k}$ in terms of spherical polar coordinates.		7	L3	CO1

Module - 3						
Q.5	a.	Find the rank of the matrix $A = \begin{bmatrix} 2 & 1 & -1 & 3 \\ 1 & 2 & 4 & 3 \\ 3 & 6 & 12 & 9 \\ 3 & 3 & 3 & 6 \end{bmatrix}$	6	L2	CO2	
	b.	Using Gauss Jordan method, solve the system of equations $x + y + z = 9, x - 2y + 3z = 8, 2x + y - z = 3$.	7	L2	CO2	
	c.	Diagonalize the matrix $\begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$	7	L3	CO2	
OR						
Q.6	a.	Investigate the value of μ and λ such that the equations, $x + y + z = 6, x + 2y + 3z = 10, x + 2y + \lambda z = \mu$, may have (i) unique solution (ii) infinite solution (iii) no solution	6	L2	CO2	
	b.	Using Gauss Elimination method, solve the system of equations $x + 2y + z = 3, 2x + 3y + 3z = 10, 3x - y + 2z = 13$	7	L2	CO2	
	c.	Find the eigen value and eigen vectors of the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$	7	L3	CO2	
Module - 4						
Q.7	a.	Determine whether the vectors $(8,0,5)$ is a linear combination of the vectors $(1,2,3), (0,1,4), (2,-1,1)$	6	L2	CO3	
	b.	Find the basis and dimension of the subspace spanned by the vectors $(2,4,2), (1,-1,0), (1,2,1), (0,3,1)$ in $V_3(R)$.	7	L2	CO3	
	c.	Find the basis and dimension of the row space, column space and null space of the matrix $\begin{bmatrix} 1 & -1 & 1 & 3 & 2 \\ 2 & -1 & 1 & 5 & 1 \\ 0 & 1 & -1 & -1 & -3 \end{bmatrix}$	7	L3	CO3	
OR						
Q.8	a.	Find the coordinates of the vector $v = (0,1,3)$ with respect to the basis $B = \{(1,1,0), (0,1,1), (1,0,1)\}$.	6	L2	CO3	
	b.	Define inner product space. Given $u = (1,2,4), v = (2,-3,5), w = (4,2,-3)$ in R^3 . Find (i) $\langle u, v \rangle$ (ii) $\langle v, w \rangle$ (iii) $\langle u, w \rangle$ (iv) $\ u\ $ (v) $\ v\ $	7	L2	CO3	
	c.	What is a subspace? Prove that the subset $W = \{(x,y,z)/x - 3y + 4z = 0\}$ of the vector space R^3 is a subspace.	7	L3	CO3	
Module - 5						
Q.9	a.	Show that the transformation $T: R^2 \rightarrow R^3$ defined by $T(x,y) = (x+y, x-y, y)$ is a linear transformation.	6	L2	CO3	

	b.	Prove that the transformation $T: R^2 \rightarrow R^2$ is singular and find its kernel, if the transformation is given by $T(x, y) = (2x + 4y, x - 2y)$.	7	L2	CO3
	c.	Verify Rank- Nullity theorem for the transformation $T: R^3 \rightarrow R^3$ defined by $T(x, y, z) = (x + 2y, y - z, x + 2z)$	7	L3	CO3
OR					
Q.10	a.	Find the matrix representing the linear transformation $T: R^3 \rightarrow R^4$ defined by $T(x, y, z) = (x + y + z, 2x + z, 2y - z, 6y)$	6	L2	CO3
	b.	If $T: R^3 \rightarrow R^3$ is a linear transformation defined by $T(x, y) = (x + z, x - z, y)$ Show that T is invertible and find T^{-1}	7	L2	CO3
	c.	Verify Rank- Nullity theorem for the transformation $T: R^3 \rightarrow R^2$ defined by $T(x, y, z) = (x + y, z)$	7	L3	CO3

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First Semester B.E/B.Tech. Degree Examination, June/July 2026
Calculus and Linear Algebra: CSE Stream

Time: 3 hrs.

Max. Marks: 100

- Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. M : Marks , L: Bloom's level , C: Course outcomes.
 3. VTU formula Hand Book is permitted.

Module - 1			M	L	C
1	a.	If $z = u^2 + v^2$ where $u = at^2$ & $v = 2at$ find $\frac{dz}{dt}$	6	L2	CO1
	b.	If $u = \log(x^3 + y^3 + z^3 - 3xyz)$ then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$	7	L2	CO1
	c.	Find the extreme values of the function $f(x,y) = x^3 + y^3 - 3x - 12y + 20$	7	L3	CO1
OR					
2	a.	If $u = (x - y, y - z, z - x)$ show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$	6	L2	CO1
	b.	If $u = x^2 + y^2 + z^2$, $v = xy + yz + zx$, $w = x + y + z$ Find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$	7	L2	CO1
	c.	Expand $e^x \cos y$ in the power of x and y as far as terms of third term using maclaurins series	7	L3	CO1
Module - 2					
3	a.	Find the directional derivative of $\phi = x^2yz + 4xz^2$ at the point $(1, -2, -1)$ along $2i - j - 2k$	6	L2	CO1
	b.	If $\vec{F} = \nabla(xy^3z^2)$ find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$ at the point $(1, -1, 1)$	7	L2	CO1
	c.	Express the vector $\vec{A} = zi - 2xj + yk$ in cylindrical coordinates	7	L3	CO1
OR					
4	a.	Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at $(2, -1, 2)$	6	L2	CO1

	b.	Show that $\vec{F} = (y+z)\mathbf{i} + (z+x)\mathbf{j} + (x+y)\mathbf{k}$ is irrotational. Also, find a scalar function ϕ Such that $\vec{F} = \nabla\phi$.	7	L	
	c.	Express $\vec{F} = y\mathbf{i} - z\mathbf{j} + x\mathbf{k}$ in spherical polar coordinates.	7	L3	
Module – 3					
5	a.	Find the rank of the matrix $A = \begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$	6	L2	CO2
	b.	Using Gauss- Jordan method, Solve the system of equation $2x + 5y + 7z = 52, 2x + y - z = 0, x + y + z = 9$	7	L2	CO2
	c.	Diagonalize the matrix $\begin{bmatrix} -19 & 7 \\ -42 & 16 \end{bmatrix}$	7	L3	CO2
OR					
6	a.	Test for consistency and solve $x + y + z = 6, x - y + 2z = 5, 3x + y + z = 8$	6	L2	CO2
	b.	Using Gauss-elimination method, solve the system of equation $x + y + z = 9, x - 2y + 3z = 8, 2x + y - z = 3$	7	L2	CO2
	c.	Find the eigen value and eigen vectors of the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$	7	L3	CO2
Module – 4					
7	a.	Determine whether the vectors $(5, 9, 5)$ is a linear combination of the vectors $(2, 1, 4), (1, -1, 3)$ and $(3, 2, 5)$.	6	L2	CO3
	b.	Determine whether $W = \{(x_1, x_2, x_3) / x_1 + x_2 + x_3 = 0\}$ is a subspace of $\mathbb{R}^3$ or not	7	L2	CO3

	c.	Find the basis and dimension of the row space, column space and nullspace of the matrix $\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$	7	L3	CO3
OR					
8	a.	Find the basis and dimension of the subspace of v spanned by the vectors $\{(1, 2, 3, 2), (2, 3, 5, 1), (1, 3, 4, 5)\}$	6	L2	CO3
	b.	Find the inner products (v_1, v_2) , (v_1, v_3) and (v_2, v_3) where $V_1 = (1, 1, 1, 1)$, $V_2 = (1, 2, 4, 5)$, $V_3 = (1, -3, -4, -2)$	7	L2	CO3
	c.	Find the coordinates of the vector $V = (1, -3, 2)$ With respect to the basis $S = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$	7	L2	CO3
Module – 5					
9	a.	If $T: V_1(R) \rightarrow V_3(R)$ is defined by $T(x) = (x, x^2, x^3)$ verify whether T is linear or not	6	L2	CO3
	b.	Verify Rank - Nullity theorem for the transformation $T: V_3(R) \rightarrow V_2(R)$ is defined by $T(x, y, z) = (y - x, y - z)$	7	L3	CO3
	c.	Prove that the transformation $F: R^2 \rightarrow R^2$ is singular and find its Kernel if the transformation $F(a, b) = (2a - 4b, 3a - 6b)$	7	L2	CO3
OR					
10	a.	If $F: V_3(R) \rightarrow V_3(R)$ is defined by $F(x, y, z) = (x + y, y + z)$ show that F is a linear transformation.	6	L2	CO3
	b.	Determine the matrix of the linear transformation $T: V_2(R) \rightarrow V_3(R)$ defined by $T(x, y) = (-x + 2y, y, 3x + 3y)$ relative to the basis.	7	L2	CO3
	c.	Verify Rank-Nullity theorem for the transformation $T: V_3(R) \rightarrow V_3(R)$ defined by $T(x, y, z) = (x + y, x - y, 2x + z)$.	7	L3	CO3
